

SECTION A [24 Marks]

1. Define each of the following terms:

[4 marks]

(a) Loop (b) Program (c) Machine language (d) Computer programming

2. Write the general syntax for each of the following:

[4 marks]

(a) do while loop (b) function prototype (c) nested for loop (d) switch case

[4 marks]

3. write a program that will print the following output on the screen:

BACHELOR OF ENGINEERING IN MECHANICAL ENGINEERING – SEC. YEAR
MECHANICAL ENGINEERING DEPARTMENT
CS 8200 COMPUTER PROGRAMMING

4. Write a simple C program that will display your full name followed by the words "I'm the MUST student" 1000 times on the screen. Make sure that in each iteration the output is displayed in different line. Use for loop.

[4 marks]

5. Differentiate each of the following terms:

a) While loop and Do while loop b) Global variable and Local variable

[4 marks]

6. Write the output of the following source code

[4 marks]

```
#include<stdio.h>
int main()
{
    float num1 = 4.6, num2 = 7.8, result;
    printf("This program compares %f and %f\n", num1, num2);
    result = (num1>num2)? num1: num2;
    printf("Where the greater number is %f\n", result);
    return 0;
}
```

SECTION B [36 Marks]

7. a) Assume you have been given three array of characters which are first_name, middle_name as well as last_name and those array were initialized based on your actual names. Write a C program that will display your first name followed by initials of your middle name and surname separated by commas.

[5 marks]

b) Write a C program that will convert either kilometer to meter or vice versa depending on user's choice. Your program should consist three functions i.e. main(), kilo_Meter() and meter_Kilo().

The main() provides available options as well as prompting user to enter the required parameters from the keyboard. The main() invokes either kilo_Meter() or meter_Kilo() depending on conversion opted by the user and display results. The kilo_Meter() converts kilometer to meter and meter_Kilo() converts meter to kilometer. Use switch case. [7 marks]

8. a) Write a C source code that will generate the following pattern. Use while. [6 marks]

```

1 2 3 4 5 6
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1

```

b) Write a program that will accept any two floating numbers from the user and calculate the sum and average. [6 marks]

9. a) Write a program that prints the numbers between 20 and 99 that can be evenly divided by 4. Also your program should compute sum of the extracted numbers. Use do while loop. [6 marks]

b) Write a C program that will prompt a user to enter any integer number from the keyboard and determine its status if the number positive or negative or zero. Let your program display the entered number while stating its status. [6 marks]

10. a) Write a C program that will extract numbers divisible by 15 between 30 and 1000 exclusively. The program should also calculate the sum the series. Use while loop. [6 marks]

b) What is the output of following source code: [6 marks]

```

#include<stdio.h>
int main()
{
    int i = 8;
    while(i>=1)
    {
        printf("%d\t\t", i);
        i--;
    }
    printf("\n\n");
    printf("The odd numbers are :\n");
    for(i=8; i>=1; i--)
    {
        if(i%2 == 1)
            printf("%d\t\t", i);
    }
    return 0;
}

```


1 a) loop, loop ~~can~~ consists of two parts, a body of loop and a control ~~se~~ statement.

b) program, contains various functions which are part of a library

c) Machine language, Also known as machine language, is the elemental language of computers.

d) Computer programming, is a high level and general purpose programming language that is ideal for developing firmware.

2. a) general syntax

do

{

statements(s)

}

While (condition)

by General syntax,

return-type function-name (parameter list)

{

body of the function

}

c) General syntax.

for (initialization; condition; increment/decrement)

{

for (initialization; condition; increment/decrement)

{

statement of the inside loop.

}

statement of the outer loop

}

3 #include <stdio.h>

int main ()

{

int a, b = 48;

for (a = 1; a <= b; a++)

printf("x");

printf("\n");

printf("BACHELOR OF ENGINEERING IN MECHANICAL
ENGINEERING - 5TH YEAR \n");

printf("MECHANICAL ENGINEERING DEPARTMENT
\n");

printf("CS 8200 COMPUTER PROGRAMMING \n");

for (a = 1; a <= b; a++)

printf("=");

printf("\n");

return 0;

}


```

4 #include <stdio.h>
   int main ()
   {
       int a, b = 1000;
       for (a = 1; a <= b; a++)
           printf("Mwiga Nchemba I'm the Must student\n");
       return 0;
   }

```

5. a) While loop, is a variant of the while statement in which the condition test is performed at the ~~to~~ top of the loop.

~~While!~~

While!

b) do while loop, is a variant of the while statement in which the condition test is performed at the bottom of the loop.

```

4 #include <stdio.h>
   int main ()
   {
       int a, b = 1000;
       for (a = 1; a <= b; a++)
           printf("Mwiqua NCHENBA I'm the Must student\n");
       return 0;
   }

```

5. a) while loop, is a variant of the while statement in which the condition test is performed at the ~~to~~ top of the loop.

~~while!~~

while!

b) do while loop, is a variant of the while statement in which the condition test is performed at the bottom of the loop.

5 by Global variable, is defined outside of the function.

While.

local variable, variable that is declared inside a function or block.

6. This program compares 4.6 and 7.8
Where the greater number is 7.8

7 a) ~~#include <stdio.h>~~

7 a) #include <stdio.h>

int main ()

{

char First_name Middle_name Last_name[] = "Elia, A, #";

printf("%s\n", First_name Middle_name Last_name);

return 0;

}

9 by @include cstdio.h

int main ()

{

float kilometer, meter, a, b, c, d;

int choice;

printf("press 1 To convert kilometer to meter");

printf("\n 2 To convert meter to kilometer\n");

scanf("%d", &choice);

switch(choice)

{

Case 1:

printf("Enter any length in kilometer\n");

scanf("%f", &a);

b = a * 1000;

printf("length in meter: %f", b);

break;

Case 2:

printf("Enter any length in meter\n");

scanf("%f", &c);

d = c / 1000;

printf("Length in meter kilometer: %f", d);

break;

default:

printf("Invalid choice");

return 0;

8. a) #include <stdio.h>

```
int main ()
```

```
{
```

```
for (int a = 6; a >= 1; a--) {
```

```
for (int b = 1; b <= a; b++) {
```

```
printf("%d", b); }
```

```
printf("\n");
```

```
printf("\n"); }
```

```
return 0;
```

```
}
```

b) #include <stdio.h>

```
int main ()
```

```
{
```

```
int num1, num2, sum, Average;
```

```
printf("Enter two numbers\n");
```

```
scanf("%d %d", &num1, &num2);
```

```
sum = num1 + num2;
```

```
Average = sum / 2;
```

```
printf("The sum and average are: %d, %d", sum, Average);
```

```
return 0;
```

```
}
```

9. a) #include <stdio.h>

int main ()

{

int a = 20, sum;

do

{

if (a % 4 == 0)

{

printf("%d", a);

}

sum = sum + a;

printf("The sum is: %d", sum);

a++;

}

while (a <= 99);

return 0;

}

q.6) #include <stdio.h>

int main ()

{

int a

printf("Enter any number");

scanf("%d", &a);

if (a > 0)

{

printf("The number is positive \n");

}

else if (a < 0)

{

printf("The number is negative \n");

}

else

{

printf("The number is zero \n");

}

return 0;

}

10 a) #include <stdio.h>

int main ()

int a=31, b=1000, sum;

while (a < 1000)

{

if (a%15 == 0)

{

printf ("%d\n", a);

}

sum = sum + a;

a++;

}

printf ("%d\n", sum);

return 0;

}

10 b) 8 7 6 5 4 3 2 1

The odd numbers are:

7 5 3 1

SECTION A [24 Marks]

1. Define each of the following as used in C++ programming:

- a) Class b) Polymorphism c) Inheritance d) Comments

[4 marks]

2. Write a simple program to demonstrate the structure of a C++ program. Let your program display your full name, registration number and year of completion your diploma studies.

Where each item should be in different line.

[4 marks]

3. Write the output of the following source code:

[4 marks]

```
#include <iostream>
using namespace std;
int main()
{
    float num1 = 4.456, num2 = 7.128, result;
    cout << "This program compares " << num1 << " and " << num2 << endl;
    result = (num1 > num2) ? num1 : num2;
    cout << "The greater number is " << result;
    return 0;
}
```

4. Differentiate each of the following:

- i) global variable and local variable ii) array and string

[4 marks]

5. Write a simple program that will calculate the following arithmetic operation and display as follows:

$$8 \times 5 = 40$$

[4 marks]

6. Write the general syntax of each of the following:

[4 marks]

- a) Switch case b) do while loop c) function definition d) function declaration

Section B [36 Marks]

7. a) Write the C++ source code that will generate the following pattern:

[6 marks]

```
1 2 3 4 5 6
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

b) Write a C++ program that prints 105 up to 1245 in descending order. Also let your program compute the sum of this series. Use do while loop.

[6 marks]

8. a) Write a C++ program that will accept student scores for eight subjects from the user. Finally, your program should display those score and determine the highest score. [6 marks]
- b) Write a C++ source code that will calculate either product or quotient of any two floating numbers depending on user's choice. Assume that the numbers are non-zero. Use switch case. [6 marks]
9. a) Write a C++ program that calculate the volume of the cylinder with radius 15cm and height 70cm. Your program should consists two functions: which are main () and veylinder (). The main function has the values of the radius as well as height and invoke veylinder () to calculate the volume of the cylinder and display the result. The veylinder () has the procedure to compute the volume of the cylinder. Note that $v = \pi r^2 h$ where r and h are radius and height of the cylinder. [8 marks]
- b) Assume you have been given three array of characters which are your first_name, middle_name as well as last_name and those names were initialized based on your full name. Write a C++ program that will display your initials separated by comma. [4 marks]
10. a) Write a C++ program that will extract numbers divisible by 15 between 30 and 1000 exclusively. The program should also calculate the sum the series. Use while loop. [6 marks]
- b) What is the output of following source code [6 marks]

```
#include <iostream>
using namespace std;
int main()
{
    int i = 8;
    lab:
        cout<<i<<"\n";
        i--;
        if(i>=1)
            goto lab;
    cout<<endl;
    cout<<"The odd numbers are : "<<endl;
    for(i=8; i>=1; i--)
    {
        if(i%2 == 1)
            cout<<i<<"\n";
    }
    return 0;
}
```

1) A class, is a way to bind the data and its associated functions together.

by polymorphism, its the ability to take more than one form.

c) Inheritance, When one object acquires all the properties and behaviours of the parent objects.

d) Comments are simply text that is ignored by the compiler, but that may inform the reader of what you are doing at any particular point in your program.

2. #include <stdio.h>

int main ()

{

char name [] = "Ally John";

int reg = 201001111001;

int year = 2002;

printf("%s\n", name);

printf("%d\n", reg);

printf("%d\n", year);

return 0;

}

3

This program compares 4.436 and 7.128

The greater number is 7.128.

i) ^{Local} ~~global~~ variable, Variable that are declared inside a function or block.

While.

global variable, are defined outside of the function, usually on top of the program.

ii) Array, is a data structure that stores a set of elements of the same data type.

While:

string, is a set of characters.

```
#include <stdio.h>
```

```
int main ()
```

```
{
```

```
int a = 5; b = 8;
```

```
int x;
```

```
x = a * b;
```

```
printf ("%d", x);
```

```
return 0;
```

```
}
```

Q. Switch case:

Syntax

```
switch (Expression)
```

```
{
```

```
    case 1:
```

```
        block 1;
```

```
        break;
```

```
    case 2:
```

```
        block 2;
```

```
        break;
```

```
    case n:
```

```
        block n;
```

```
        break;
```

```
    default:
```

```
        statement;
```

Q. do while loop:

Syntax

```
do
```

```
{
```

```
    body of the loop
```

```
}
```

```
while (condition);
```

6 c) Function definition
syntax

return-type function-name(parameter list).

d) Function declaration
syntax

type function-name (type [argname] [, type-...])

7 a) #include <stdio.h>

int main ()

{

for (int a = 6; a >= 1; a--) {

for (int b = 1; b <= a; b++) {

printf("%d", b); }

printf("\n"); }

return 0;

}


```
7 b) #include <stdio.h>
```

```
int main ()
```

```
{
```

```
int a = 1045, sum;
```

```
do
```

```
{
```

```
sum = sum + a;
```

```
a--;
```

```
printf("The sum of series is: %d", sum);
```

```
}
```

```
while (a >= 105);
```

```
return 0;
```

```
}
```

8 a) #include <stdio.h>

int main ()

{

int a[8], size, i, big;

printf("Enter size of the array \n");

scanf("%d", &size);

for (i=0; i<=size; i++)

{

scanf("%d", &a[i]);

}

big = a[0];

for (i=1; i<=size; i++) {

if (big < a[i])

big = a[i];

}

printf("\n The biggest element is %d", big);

return 0;

}

```
8 by #include <stdio.h>
```

```
int main ()
```

```
{
```

```
int num1, num2;
```

```
int choice;
```

```
printf("press \n 1 To calculate product of two number-  
");
```

```
printf("\n 2 To calculate the quotient of the two num-  
ber \n");
```

```
scanf("%d", &choice);
```

```
printf("Enter any two number \n");
```

```
scanf("%d %d", &num1, &num2);
```

```
switch (choice) {
```

```
case 1:
```

```
int a;
```

```
a = num1 * num2;
```

```
printf("%d \n", a);
```

```
break;
```

```
case 2: -
```

```
int b
```

```
b = num1 / num2;
```

```
printf("%d \n", b);
```

```
break;
```


8 continue - - - -

default:

```
printf("Invalid choice");
```

```
}
```

```
return 0;
```

```
}
```

9 a) #include <stdio.h>

float vcylinder(int r, int h);

$\pi = 3.14;$

int main ()

{

int radius = 15, height = 70;

float volume;

volume = vcylinder (radius, height);

printf("Volume is %d\n", volume);

return 0;

}

float vcylinder(int r, int h)

{

float v;

$v = \pi * r * r * h;$

return v;

}

10 a) #include <stdio.h>

```
int main ( )
```

```
int a=31, b=1000, sum;
```

```
while (a < 1000)
```

```
{
```

```
if (a%15 == 0)
```

```
{
```

```
printf("%d\n", a);
```

```
}
```

```
sum = sum + a;
```

```
a++;
```

```
}
```

```
printf("%d\n", sum);
```

```
return 0;
```

```
}
```

10 b)

8 7 6 5 4 3 2 1

The odd numbers are:

7 5 3 1